

EE 473 - HOMEWORK 1

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1) a) $x_a(t) = 3 \cos(5t + \pi/6)$

Signal is periodic with $T_a = \frac{2\pi}{5}$

* $\sin(\cdot)$ and $\cos(\cdot)$ functions are periodic with 2π

b) $x(n) = 3 \cos(5n + \pi/6)$

↳ This is similar to the previous option but this time signal is discrete. Since $\frac{2\pi}{5}$ is not an integer and any integer multiple of 5 can't be equal to any integer multiple of 2π (due to π being irrational); signal is NOT periodic.

c) $x(n) = 2 \exp[j(n/6 - \pi)] = 2 \cos(n/6 - \pi) + 2j \sin(n/6 - \pi)$

↳ As in the last case, any integer multiple $\frac{1}{6}$ can't be equal to $2\pi k$ ($k \in \mathbb{Z}$). Hence, signal is NOT periodic.

d) $x(n) = \cos(n/8) \cos(\pi n/8)$

↳ $\cos(\pi n/8)$ is periodic with $N=16$ but $\cos(n/8)$ is not periodic due to same reasons of previous 2 options. Thus, overall signal is NOT periodic.

e) $x(n) = \cos(\pi n/2) - \sin(\pi n/8) + 3 \cos(\pi n/4 + \pi/3)$

↳ $\cos(\pi n/2)$ is periodic with $N_1 = 4$

↳ $\sin(\pi n/8)$ is periodic with $N_2 = 16$

↳ $\cos(\pi n/4 + \pi/3)$ is periodic with $N_3 = 8$

→ Resulting signal is periodic with period equal to the least common multiple of individual periods which is $N = 32$

$$2) \quad y(n) = (\delta(n) + h_1(n)) * x(n) * h_2(n)$$

$$\text{where } h_1(n) = \beta \delta(n-1), \quad h_2(n) = \alpha^n u(n)$$

$$\begin{aligned} a) \quad y(n) &= [(\delta(n) + h_1(n)) * h_2(n)] * x(n) \\ &= (\delta(n) * h_2(n) + \underbrace{h_1(n) * h_2(n)}_{\beta \delta(n-1)}) * x(n) = [h_2(n) + \beta h_2(n-1)] * x(n) \\ &\quad \xrightarrow{\text{Utilized } x(n) * \delta(n) = x(n)} \\ &= [\alpha^n u(n) + \beta \alpha^{n-1} u(n-1)] * x(n) \end{aligned}$$

$$\boxed{h(n) = \alpha^n u(n) + \beta \alpha^{n-1} u(n-1) = \delta(n) + \alpha^n \left(1 + \frac{\beta}{\alpha}\right) u(n-1)}$$

$$\begin{aligned} b) \quad \text{Sol 1} \quad y(n) &= (\alpha^n u(n) + \beta \alpha^{n-1} u(n-1)) * x(n) \\ \alpha y(n-1) &= (\alpha^n u(n-1) + \beta \alpha^{n-1} u(n-2)) * x(n) \end{aligned} \quad \begin{array}{l} \text{Since LTI system has time} \\ \text{invariance } y(n-1) = h(n-1) * x(n) \end{array}$$

$$\begin{aligned} y(n) - \alpha y(n-1) &= \alpha^n \underbrace{(u(n) - u(n-1))}_{\delta(n)} * x(n) + \beta \alpha^{n-1} \underbrace{(u(n-1) - u(n-2))}_{\delta(n-1)} * x(n) \\ &= \alpha^0 \delta(n) * x(n) + \beta \alpha^{1-1} \delta(n-1) * x(n) \\ &= x(n) + \beta x(n-1) \end{aligned}$$

$$\boxed{y(n) - \alpha y(n-1) = x(n) + \beta x(n-1)}$$

Sol 2 Take the Z-transform of both sides:

$$\mathcal{Z}\{y(n)\} = Y(z) = \mathcal{Z}\{\alpha^n u(n) + \beta \alpha^{n-1} u(n-1)\} \mathcal{Z}\{x(n)\} \quad \begin{array}{l} \text{utilized the property} \\ \mathcal{Z}\{x_1 * x_2\} = X_1(z) X_2(z) \end{array}$$

$$\mathcal{Z}\{\alpha^n u(n)\} = \sum_{n=-\infty}^{\infty} \alpha^n u(n) z^{-n} = \sum_{n=0}^{\infty} \left(\frac{\alpha}{z}\right)^n = \frac{1}{1 - \alpha z^{-1}} \quad \text{for } |z| > \alpha$$

$$\mathcal{Z}\{\alpha^{n-1} u(n-1)\} = z^{-1} \mathcal{Z}\{\alpha^n u(n)\} = \frac{z^{-1}}{1 - \alpha z^{-1}} \quad \text{for } |z| > \alpha$$

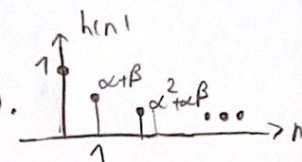
2b-continued)

$$Y(z) = \underbrace{\left[\frac{1}{1-\alpha z^{-1}} + \frac{\beta z^{-1}}{1-\alpha z^{-1}} \right]}_{H(z)} X(z) \Rightarrow Y(z) - \alpha z^{-1} Y(z) = X(z) + \beta z^{-1} X(z)$$

Taking the inverse $\mathbb{Z}$ -transform and using time shift property $[x(n-1) \xrightarrow{\mathbb{Z}} z^{-1} X(z)]$

$$\boxed{y(n) - \alpha y(n-1) = x(n) + \beta x(n-1)}$$

c) $h(n) = \alpha^n u(n) + \beta \alpha^{n-1} u(n-1)$ is 0 for $n < 0$.



Therefore system is causal (impulse response is right sided)

→ Looking at the transfer function $H(z) = \frac{1+\beta z^{-1}}{1-\alpha z^{-1}}$, we see that there exist a pole at $z=\alpha$

→ Since system is causal ROC reaches outward of the pole: $|z| > |\alpha|$

→ For system to be stable ROC must contain the unit circle $|z|=1$ and thus, we should have $|\alpha| < 1$ for stability.

3) $y(n) = x(n) + 0.4 y(n-1)$ with $x(n) = \sin(0.2\pi n)$

a) Plots are on the next page

b) $Y(z) = \frac{X(z)}{1-0.4z^{-1}} \Rightarrow$ Since TF has a pole, digital filter is IIR

c) We know that when sinusoids (complex exponentials in general) are input to an LTI system, steady-state output shifted and scaled version of the exponential. Further, amount of shift and scaling is determined by the TF when $z = e^{j\omega}$

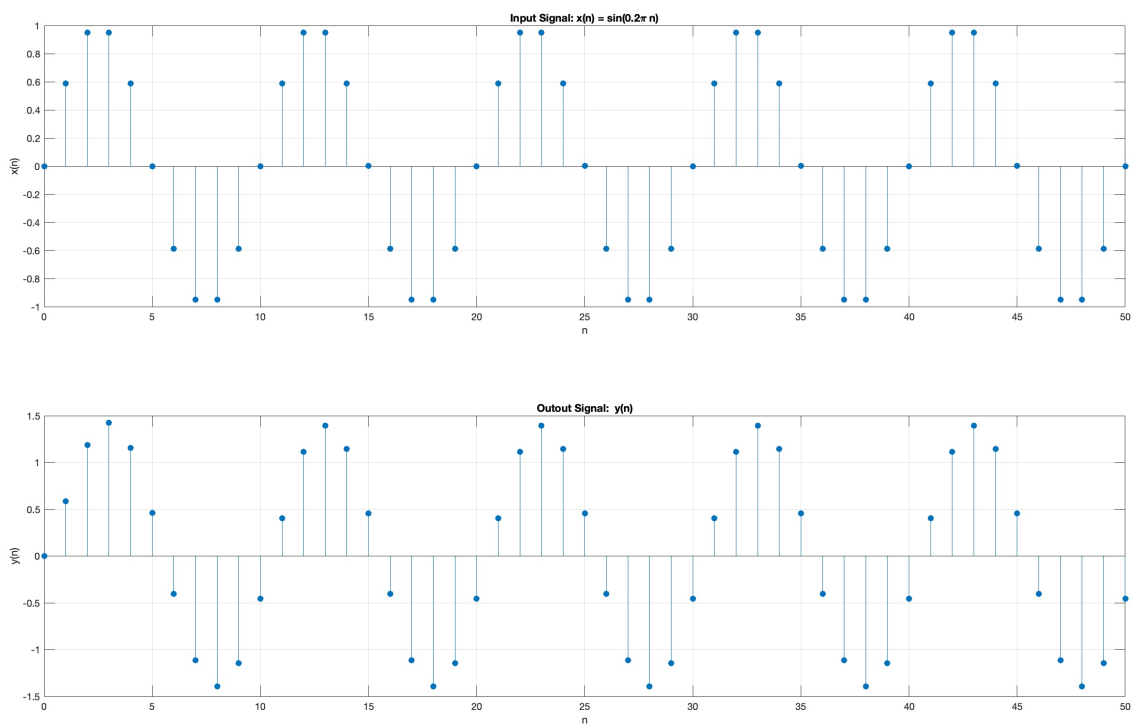
$$e^{j\omega_0 n} \rightarrow \boxed{\text{LTI}} \rightarrow |H| e^{j(\omega_0 n + \angle H)} \Rightarrow y(n) = |H|_{\omega=\omega_0} \sin(0.2\pi n + \angle H_{\omega=\omega_0})$$

$$H(\omega_0) = \frac{1}{1-0.4e^{j\omega_0}} \Rightarrow |H(\omega)|_{\omega=0.2\pi} = \frac{1}{[1-\cos(0.2\pi) \times 0.4]^2 + [0.4 \sin(0.2\pi)]^2}^{1/2} \approx 1.396$$

$$\angle H(\omega)_{\omega=0.2\pi} = -\arctan\left[\frac{0.4 \sin(0.2\pi)}{1-0.4 \cos(0.2\pi)}\right] = -\arctan(0.3476) = -0.334$$

$$y(n) \approx 1.396 \sin(0.2\pi n - 0.334)$$

Input and output signal of question 3, plotted for 5 periods



4) $h(n) = e^{-n} u(n) \Rightarrow \text{LTI}$

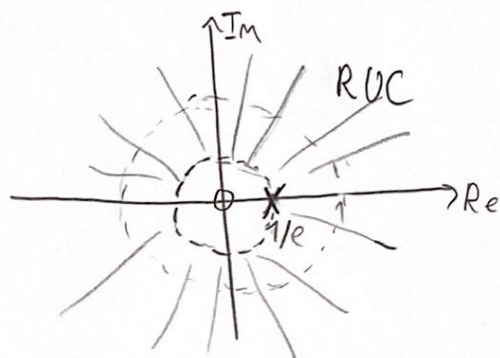
a) System is causal since $h(n) = 0$ for $n < 0$

b) System is stable since $\sum_{n=-\infty}^{\infty} |e^{-n} u(n)| = \sum_{n=0}^{\infty} e^{-n} = \frac{1}{1-\frac{1}{e}} < \infty$

c) $H(z) = \sum_{n=-\infty}^{\infty} e^{-n} u(n) z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{e} z\right)^n = \frac{1}{1-\frac{1}{e} z^{-1}} = \frac{z}{z-\frac{1}{e}} \quad \text{for } |z| > \frac{1}{e}$

d) $Y(z) = \frac{X(z)}{1-\frac{1}{e} z^{-1}} \Rightarrow Y(z) - \frac{1}{e} z^{-1} Y(z) = X(z)$

$y(n) - \frac{1}{e} y(n-1) = x(n)$



e) $y(n) = h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(n-k) x(k) = \sum_{k=-\infty}^{\infty} e^{-(n-k)} u(n-k) x(k)$

$y(n) = \sum_{k=-\infty}^n x(k) e^{-n-k}$

5) $x(n) = s(n) + r_1 s(n-k_1)$

a) $r_{xx}(l) = \sum_n x(n+l) x(n) = \sum_n (s(n+l) + r_1 s(n+l-k_1)) (s(n) + r_1 s(n-k_1))$
 $= \underbrace{\sum_n s(n+l) s(n)}_{r_{ss}(l)} + r_1 \underbrace{\sum_n s(n+l) s(n-k_1)}_{r_{ss}(l+k_1)} + r_1 \underbrace{\sum_n s(n+l-k_1) s(n)}_{r_{ss}(l-k_1)} + r_1^2 \underbrace{\sum_n s(n+l-k_1) s(n-k_1)}_{r_{ss}(l)}$

$r_{xx}(l) = (1+r_1^2) r_{ss}(l) + r_1 [r_{ss}(l+k_1) + r_{ss}(l-k_1)]$

b) Even if we know the autocorrelation of $s(n)$, $r_{ss}(l)$; we still have 2 unknowns with 1 eq. Thus, it is NOT possible to obtain r_1, k_1 just by observing $r_{xx}(l)$ without any assumptions on $s(n)$.